

When More Data Make the Wrong Model More Certain

Network Dependence, Observation Grain, and Recoverability
in Agent-Based SIR Models

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Abstract

Pandemic transmission depends on contact network structure. The mean-field ODE SIR model is therefore strictly misspecified as a mechanistic account: the SI/N term encodes uniform mixing, not actual contact topology. But misspecification alone does not determine whether the network mechanism is empirically recoverable. The central question is whether the observation process—the grain at which infections are measured and the clock at which they are reported—contains enough information to identify network structure.

We demonstrate that under weekly aggregate incidence, it does not. Data are generated from a network SIR process with known community structure; a mean-field ODE is calibrated to those observations. The ODE posterior concentrates near a pseudo-true parameter that absorbs network structure into a lower effective aggregate rate, understating the counterfactual reference by 23% in our baseline experiment. Collecting more independent outbreaks at the same grain and clock reduces posterior width fourfold without reducing bias. More data make the wrong model more certain when the additional data are collected under the same insufficient observation process. We connect this to PAC-Bayesian regularization: the ABM must allow heterogeneity, but the data—given their grain and clock—determine whether that heterogeneity is supported or should remain uncertain.

1 The Problem

Agent-based modelers routinely simulate epidemic spread, opinion dynamics, and market behavior at the level of individual agents [Epstein, 2008]. The simulation produces rich internal trajectories: who interacted with whom, in what order, under what conditions. But the data available for calibration are almost always coarser. A modeler fitting an epidemic model to weekly case reports compares a fine-grained stochastic process to aggregate counts measured at a weekly clock.

This mismatch creates a problem that is easy to overlook. When the model is calibrated to coarse data, estimated parameters may be fitting the observation process as much as the underlying mechanism. If the observation process discards information that the mechanism depends on, fitted parameters will not recover the true microscopic rates—no matter how much data are collected.

Because network dependence is assumed to be mechanistically relevant, the central inferential problem is not whether the mean-field model is exactly true. It is not. The problem is whether the observation process is rich enough to distinguish network-dependent transmission from a lower-dimensional mean-field approximation.

2 Contribution

We study when network structure in an agent-based epidemic model is empirically recoverable from observed epidemic data. We assume that pandemic transmission is network-dependent, so the mean-field SIR model is strictly misspecified as a mechanistic account. However, misspecification alone does not determine whether the network mechanism can be learned. Recoverability depends on the observation process: the grain at which infections are measured and the clock at which they are reported.

We use simulated network SIR epidemics to vary observational grain and clock. Under rich observations, network structure can be recovered using regularized inference. Under coarse aggregate observations, the same latent process supports only a reduced mean-field approximation, and additional data at the same grain make that approximation more certain without recovering the missing network mechanism. We connect this to PAC-Bayesian regularization: the ABM must allow heterogeneity, but the data determine whether that heterogeneity is supported or should remain uncertain.

We formalize this by treating agent-based models as generative statistical models with two components:

$$\text{simulator} \rightarrow \text{latent social process} \rightarrow \text{observation process} \rightarrow \text{data}.$$

The simulator encodes a theory of how agents interact. The observation process encodes how those interactions are measured—at what resolution, with what reporting rate, and with what aggregation structure. A simulation is not yet a statistical model until both components are specified.

Observational Grain and Clock

We distinguish observational grain from observational clock. Grain refers to the level at which the latent process is measured: individual infection events, group-level incidence, network-edge

exposure histories, or aggregate population counts. Clock refers to the temporal resolution at which those measurements are reported: daily, weekly, event-time, or irregular reporting intervals. In the baseline experiment, the observation process maps a continuous-time network epidemic into weekly aggregate incidence counts. This discards both network position and within-week timing.

3 The SIR Example and the Social Assumption

The standard mean-field ODE SIR model [Kermack and McKendrick, 1927, Keeling and Rohani, 2008] is:

$$\frac{dS}{dt} = -\beta_{ODE} \frac{SI}{N}, \quad \frac{dI}{dt} = \beta_{ODE} \frac{SI}{N} - \gamma I, \quad \frac{dR}{dt} = \gamma I.$$

The term SI/N is often treated as algebra. It is not. It encodes a strong social assumption: susceptible-infected contact opportunities are proportional to the product of aggregate counts. This is uniform mixing. *The product term is a social assumption.* Mean-field SIR is not wrong because it is mathematically crude; it is wrong because it encodes a social theory of uniform contact.

4 Network SIR and SI Edges

A network SIR model replaces uniform mixing with an explicit contact structure [Keeling and Rohani, 2008, Kiss et al., 2017]. For susceptible agent i with neighbors $\mathcal{N}(i)$ in contact network A , the infection probability over a short interval follows from the competing-hazards structure of independent Poisson contact processes [Kiss et al., 2017, Ch. 3]:

$$P(i \text{ infected in } [t, t + \Delta t]) = 1 - \exp\left(-\beta \Delta t \sum_{j \in \mathcal{N}(i)} \mathbf{1}\{X_j(t) = I\}\right).$$

Here β is the per-contact, per-infected-neighbor transmission rate—not the same parameter as β_{ODE} . Aggregate incidence depends on $E_{SI}(t)$, the number of susceptible-infected edges [Newman, 2002], not directly on the product SI .

A critical point about scale: β and β_{ODE} are not directly comparable. Under *uniform* mixing with mean degree $\bar{d}$, each susceptible has on average $\bar{d} \cdot I/N$ infected contacts, so the aggregate infection rate is $\beta \cdot \bar{d} \cdot SI/N$. An ODE fitted to data from such a network should therefore recover:

$$\beta_{mf,true} = \beta \cdot \bar{d}.$$

$\beta_{mf,true}$ is not a microscopic ground truth. It is a calibration benchmark: the ODE transmission rate recovered when the same per-contact process is run on a uniform network with the same mean degree and observed through the same reporting scheme. We validate this benchmark

empirically: fitting the ODE to data from an actual uniform network with mean degree $\bar{d} \approx 12$ recovers β_{ODE} within 2% of $\beta_{mf,true}$.

Under community structure, the approximation $E_{SI} \approx c SI$ fails. The epidemic burns rapidly through the seed group, then stalls at group boundaries before spreading slowly across sparse between-group edges. This two-phase dynamics produces an aggregate incidence curve that no single- β_{ODE} ODE can match exactly. The ODE instead concentrates its posterior around a pseudo-true β_{ODE}^* that best approximates the overall pacing of the aggregate curve.

The bias arises from fitting a structurally misspecified mean-field model to data whose observation process has discarded the network information needed to distinguish community transmission from a lower effective aggregate rate. The sources of failure are distinct: (1) structural misspecification (the ODE assumes uniform mixing); (2) grain loss (aggregate counts discard group and network position); (3) clock loss (weekly reporting discards within-week timing); and (4) parameter translation (β in the network model is not β_{ODE}). The gap $\beta_{ODE}^* - \beta_{mf,true}$ reflects community structure operating through all four channels.

5 Regularization and the Uniform Graph

We generate contact networks from a stochastic block model [Holland et al., 1983]:

$$\text{logit } P(A_{ij} = 1) = \alpha + \eta B_{ij},$$

where $B_{ij} = 1$ if agents i, j share a group and $B_{ij} = -1/(K - 1)$ otherwise. The parameter η controls community structure; α controls overall density. The regularization penalty is $\lambda\eta^2$.

At $\eta = 0$, all dyads have equal edge probability $\sigma(\alpha)$: the graph is Erdős–Rényi and mixing is uniform. The uniform graph is not the neutral case. It is the maximally regularized case: the network that remains when all group-based distinctions have been shrunk to zero. The standard ODE is correspondingly the maximally regularized aggregate limit. It is not what happens when an ABM gets large. It is what happens when the ABM gets large, coarse, and socially uniform.

The role of regularization is not to eliminate heterogeneity from the ABM. Heterogeneity is part of the model’s theoretical content: agents differ in contact neighborhoods, and epidemic transmission depends on the resulting susceptible-infected edges. Rather, regularization determines which forms of heterogeneity are supported by the data. In the stochastic block model, η parameterizes group-based network structure, while the penalty $\lambda\eta^2$ shrinks unsupported group structure toward the uniform graph. Thus the uniform graph is not treated as substantively true; it is the limiting case obtained when the data do not justify retaining group-level contact heterogeneity.

6 Simulation Results

Setup. We simulate $N = 1000$ agents in $K = 4$ equal groups. The community network uses $\alpha = -4.4$, $\eta = 2.5$ (mean degree ≈ 37 , within-group edge share ≈ 0.89). True parameters: $\beta = 0.04 \text{ day}^{-1}$ (per contact), $\gamma = 0.10 \text{ day}^{-1}$ (mean infectious period 10 days). The mean-field reference rate is $\beta_{mf,true} = \beta \bar{d} \approx 1.47$. We simulate 150 days, observe weekly aggregate incidence (21 time points), and fit the ODE via grid posterior with γ fixed. Let Y_k denote the observed new-infection count in week k and $\hat{Y}_k$ the corresponding ODE-predicted count; the likelihood is:

$$\log(1 + Y_k) \sim \mathcal{N}(\log(1 + \hat{Y}_k), \sigma^2).$$

Bias result. The ODE posterior concentrates near $\beta_{ODE}^* \approx 1.13$, well below the counterfactual reference $\beta_{mf,true} \approx 1.47$ —the value the ODE *would* recover from a uniform network of the same density. The bias is -0.33 , or 23% of the reference. The community epidemic’s two-phase shape (rapid within-group spread, slow between-group spread) cannot be reproduced by any single- β_{ODE} ODE; the posterior instead concentrates on the β_{ODE} that minimizes aggregate misfit.

Standard posterior concentration and acceptable aggregate fit can coexist with mechanistic failure. The fitted ODE may track weekly incidence while its parameter absorbs unobserved network structure into a biased effective rate.

More data, same grain. We generate R independent community-network outbreaks and pool their likelihoods. As R increases from 1 to 20, the posterior 95% credible interval width shrinks from 0.136 to 0.036—a fourfold reduction. The posterior MAP stays in the range $[1.11, 1.22]$, always well below $\beta_{mf,true} \approx 1.47$.

More data at the same observational grain and clock do not recover the missing mechanism. They estimate the wrong reduced model more precisely. Variance shrinks. Bias remains.

Regularization path. Varying η from 0 to 3 and pooling 20 independent outbreaks per level to obtain stable pseudo-true estimates, ODE bias becomes increasingly negative as community structure strengthens: -0.044 (9% of reference) at $\eta = 0$ and -0.48 (23% of reference) at $\eta = 3$. Absolute bias increases convexly: modest for small η and accelerating as community structure becomes strong. As community structure strengthens, the epidemic increasingly spreads in group-wise waves; the ODE compensates by fitting a lower effective transmission rate.

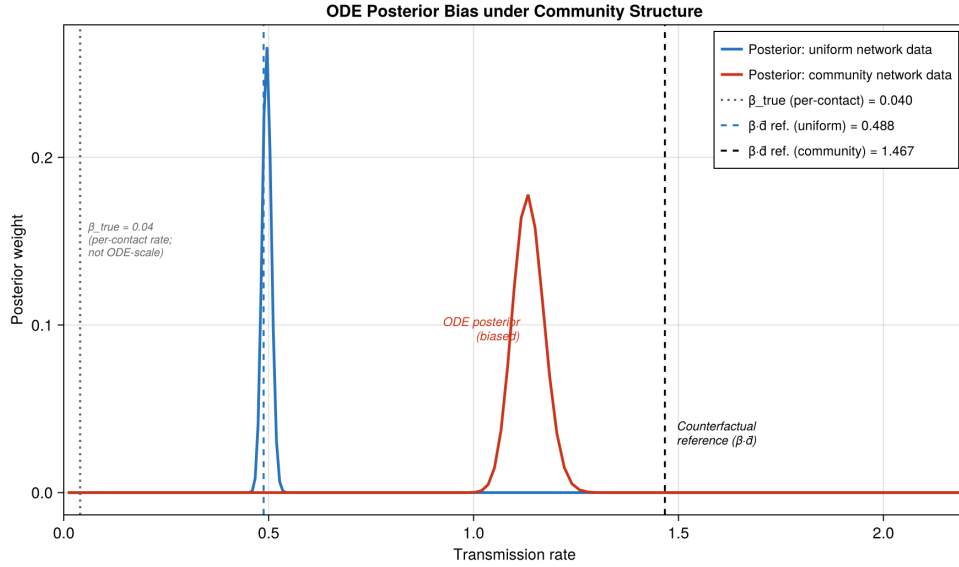


Figure 1: ODE posteriors under uniform-network data (blue, “Uniform network posterior”) and community-network data (red, “Community network posterior”). The blue posterior sits near its reference $\beta_{mf,true}$ (dashed blue line): this is what an unbiased ODE recovers from a uniform contact structure. The red posterior concentrates well to the left of $\beta_{mf,true}$ (dashed black line): community structure causes the two-phase epidemic shape that no single- β_{ODE} ODE can match, so the posterior is pulled toward a lower effective rate. Both posteriors are tight—the bias is misspecification, not uncertainty.

7 PAC-Bayesian Inference and Recoverability

In a PAC-Bayesian formulation [McAllester, 1999], candidate network structures or network-generating parameters are evaluated by predictive performance penalized by departure from a prior over simpler or more regular mechanisms. Under rich observation—such as individual infection times or sufficiently fine group-level incidence—the posterior should concentrate near the correct network structure or community parameter. Under coarse observation—such as weekly aggregate incidence—the posterior should remain diffuse over network structures or concentrate only on reduced effective parameters.

PAC-Bayes lets the model recover structure when the data support it and quantify uncertainty when they do not. This is the appropriate inferential logic for agent-based models: retain heterogeneity when the data support it, penalize it when they do not, and quantify uncertainty when the observation process cannot decide. A mechanism cannot be validated at a resolution where its identifying information has been discarded. More data help only when they add information relevant to the mechanism, not merely more replications of the same reduced observation.

This framing clarifies the role of priors. A prior is a form of regularization: it encodes

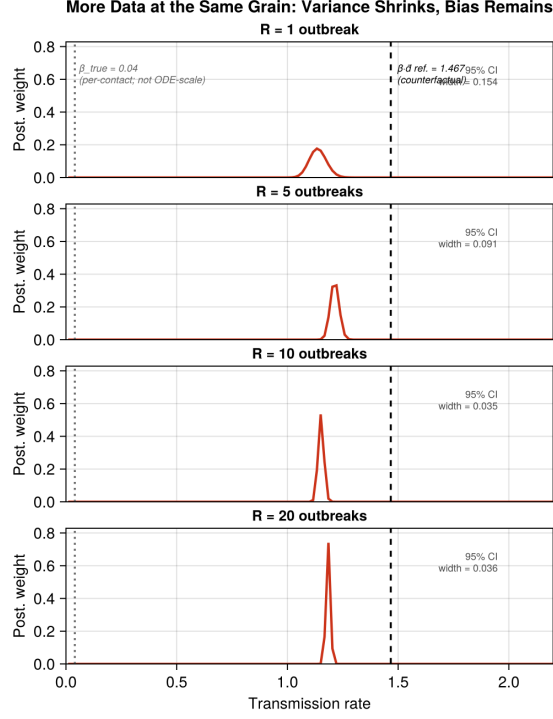


Figure 2: Four stacked panels showing the ODE posterior for β_{ODE} as the number of independent community-network outbreaks increases ($R = 1, 5, 10, 20$, top to bottom). All panels share the same horizontal axis. All additional outbreaks are observed at the same grain (weekly aggregate incidence) and the same clock (weekly reporting). The dotted vertical line marks $\beta_{true} = 0.04$ (per-contact rate; a different quantity than β_{ODE}). The dashed vertical line marks the counterfactual reference $\beta_{mf,true} \approx 1.47$ (what an unbiased ODE would recover from a uniform network of the same density). The 95% credible interval width shrinks fourfold from $R = 1$ to $R = 20$; the posterior mass does not move toward the reference. More data at the same observational grain and clock estimate the wrong reduced model more precisely.

which model configurations are penalized in the absence of data support, exactly as a penalty term shrinks parameters toward a simpler structure [Tibshirani, 1996]. Priors and perturbation schemes are in this sense dual—both express assumptions about which departures from a baseline model should require justification from the data [Hazan et al., 2016]. The practical consequence is that disputes over which prior is *correct* are largely unproductive. No prior is uniquely correct; each encodes a different regularization. What is productive is comparing priors: if conclusions are stable across a range of reasonable priors, they are robust to that modeling choice. If they are not, the instability is itself informative about what the data can and cannot identify [Gelman and Shalizi, 2013].

A practical workflow follows: (1) specify the latent process and observation process separately; (2) compare richer simulations against their mean-field limits; (3) penalize unnecessary heterogeneity in agents and network structure; (4) vary observation grain and clock; (5) examine

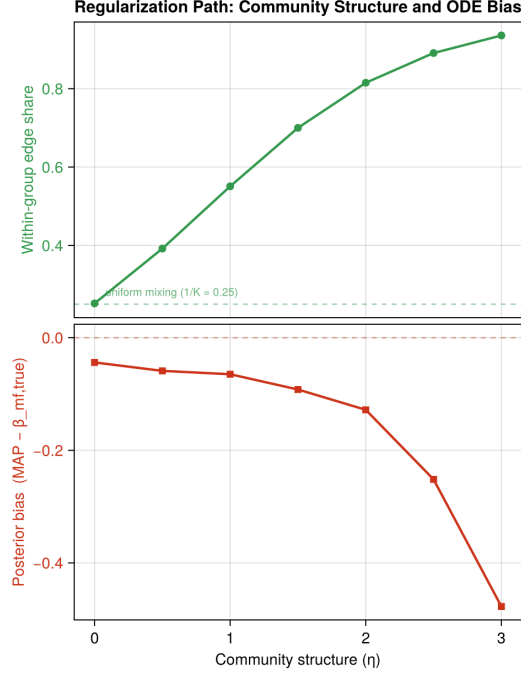


Figure 3: Regularization path, shown as two panels. *Top*: within-group edge share (green) rises from uniform mixing ($1/K = 0.25$, dashed) toward 1 as η increases. *Bottom*: ODE posterior bias ($\text{MAP} - \beta_{mf,true}$, red) becomes increasingly negative; absolute bias increases convexly with η . At $\eta = 0$ the network is approximately Erdős–Rényi and bias is near zero; at $\eta = 3$ bias reaches -0.48 .

posterior bias; (6) run posterior predictive checks.

8 Conclusion

Mean-field SIR is not wrong because it is mathematically crude; it is wrong because it encodes a social theory of uniform contact. Whether that wrongness matters empirically depends on the observation process. Fine-grained data can support network recovery. Coarse data may only support a reduced aggregate model. PAC-Bayesian regularization provides a way to distinguish these cases: retain heterogeneity when the data support it, penalize it when they do not, and quantify uncertainty when the observation process cannot decide.

Under weekly aggregate incidence, additional outbreaks reduce posterior variance around a pseudo-true ODE parameter but do not recover the network mechanism. More data at the same grain and clock estimate the wrong reduced model more precisely. The fully qualified claim is this: more data make the wrong model more certain when the additional data are collected under the same insufficient observation process. Richer grain or finer clock data can recover network structure; coarse data require regularized uncertainty rather than false mechanistic confidence.

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